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G H Raisoni International Skill Tech University, Pune

S.Y. B.Tech
(Sem-IV)
CAE III- SUMMER 2026 (2024 Pattern)
Operating Systems (25BTPCL403)

(Time: - 01 Hour) (Max. Marks: 20)

Instructions to the students:

i) Solve Q.1 or 2 and Q3 or 4. ii) Neat diagrams must be drawn wherever necessary.
iii) Figures to the right indicate full marks. iv) Use of scientific calculator is allowed
v) Assume suitable data, if necessary.

Q.No	Question	Marks	CO	BL
Q1.(a)	List different services of operating system and explain each service	[5]	I	2
b)	What is the difference between User Mode and Kernel Mode?	[5]	I	2
	OR			
Q 2.(a)	Explain Time Sharing Operating System. How does it improve CPU utilization?	[5]	I	2
(b)	Explain operating system design and implementation	[5]	I	2
Q 3.(a)	Explain Inter-process Communication.	[5]	III	2
(b)	Explain Dining Philosophers Problem.	[5]	III	2
	OR			
Q. 4 (a)	Explain the concept of semaphores in detail, along with their types.	[5]	III	2
(b)	What is Mutual Exclusion?	[5]	III	2